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Kubernetes Scheduler | Kubernetes

Kubernetes SchedulerIn Kubernetes, scheduling refers to making sure that Pods are matched to Nodes so that

Kubelet can run them.Scheduling overviewA scheduler watches for newly created Pods that have no Node assigned. For

every Pod that the scheduler discovers, the scheduler becomes responsible for finding the best Node for that Pod to run on. The scheduler reaches this placement decision taking into account the scheduling principles described below.If you want to understand why Pods are placed onto a particular Node, or if you're planning to implement a custom scheduler yourself, this page will help you learn about scheduling.kube-schedulerkube-scheduler is the default scheduler for Kubernetes and runs as part of the control plane.

kube-scheduler is designed so that, if you want and need to, you can write your own scheduling component and use that instead.Kube-scheduler selects an optimal node to run newly created or not yet

scheduled (unscheduled) pods. Since containers in pods - and pods themselves - can have different requirements, the scheduler filters out any nodes that don't meet a Pod's specific scheduling needs. Alternatively, the API lets you specify a node for a Pod when you create it, but this is unusual and is only done in special cases.In a cluster, Nodes that meet the scheduling requirements for a Pod are called feasible nodes. If none of the nodes are suitable, the pod remains unscheduled until the scheduler is able to place it.The scheduler finds feasible Nodes for a Pod and then runs a set of

functions to score the feasible Nodes and picks a Node with the highest score among the feasible ones to run the Pod. The scheduler then notifies the API server about this decision in a process called binding.Factors that need to be taken into account for scheduling decisions include

individual and collective resource requirements, hardware / software / policy constraints, affinity and anti-affinity specifications, data

locality, inter-workload interference, and so on.Node selection in kube-schedulerkube-scheduler selects a node for the pod in a 2-step operation:FilteringScoringThe filtering step finds the set of Nodes where it's feasible to

schedule the Pod. For example, the PodFitsResources filter checks whether a candidate Node has enough available resources to meet a Pod's specific resource requests. After this step, the node list contains any suitable Nodes; often, there will be more than one. If the list is empty, that

Pod isn't (yet) schedulable.In the scoring step, the scheduler ranks the remaining nodes to choose the most suitable Pod placement. The scheduler assigns a score to each Node that survived filtering, basing this score on the active scoring rules.Finally, kube-scheduler assigns the Pod to the Node with the highest ranking.

If there is more than one node with equal scores, kube-scheduler selects one of these at random.There are two supported ways to configure the filtering and scoring behavior of the scheduler:Scheduling Policies allow you to configure Predicates for filtering and Priorities for scoring.Scheduling Profiles allow you to configure Plugins that implement different scheduling stages, including: QueueSort, Filter, Score, Bind, Reserve, Permit, and others. You can also configure the kube-scheduler to run different profiles.What's nextRead about scheduler performance tuningRead about Pod topology spread constraintsRead the reference documentation for kube-schedulerRead the kube-scheduler config (v1) referenceLearn about configuring multiple schedulersLearn about topology management policiesLearn about Pod OverheadLearn about scheduling of Pods that use volumes in:Volume Topology SupportStorage Capacity TrackingNode-specific Volume LimitsFeedbackWas this page helpful?Yes

NoThanks for the feedback. If you have a specific, answerable question about how to use Kubernetes, ask it on

Stack Overflow.

Open an issue in the GitHub Repository if you want to report a problem

or

suggest an improvement. Last modified February 16, 2024 at 5:27 PM PST: Fix spelling mistake in kube scheduler (33dcba8aa6)